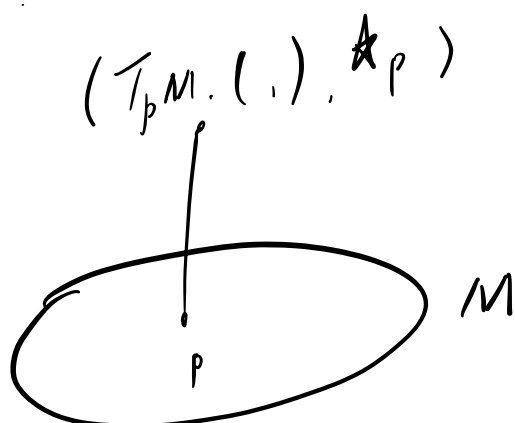


Reference : B. Dubrovin

" Painlevé transcendents "

Lecture 3

Last time :



Part I Semisimplicity.

Thm Let B be a Frobenius algebra N be its nil radical
then there exists an orthogonal decomposition

$$B \cong N \oplus M_1 \oplus \dots \oplus M_r$$

where $M_i \cong \mathbb{C} e_i$ subalgebras with e_i idempotents

rmk1 this is a special case of the Wedderburn - Artin theorem

rmk2 B is semisimple $\Leftrightarrow N(B) = 0$

rmk3 semisimplicity is an open condition. (limit of nilpotent is still nilpotent)

Def A Frobenius manifold M is called semisimple if
 $T_t M$ are semisimple for generic $t \in M$

prop if $L_B \in \text{Aut}(B)$ has distinct eigenvalues. for
 some ψ , then B is semisimple.

Pf: $B = \mathbb{C} e_i$

$$(\psi e_i, e_j) = \lambda_i (e_i, e_j) = (e_i, \psi e_j) = \lambda_j (e_i, e_j)$$

$\Rightarrow$ orthogonal.

$$\psi e_i^2 = \lambda_i e_i^2 \Rightarrow \mathbb{C} e_i = \mathbb{C} e_i^2 \text{ if } e_i^2 \neq 0$$

but $e_i^2 \neq 0$ by non-degeneracy. $\Rightarrow \mathbb{C} e_i = \mathbb{C} e_i^2$

modify e_i to idempotents.

Example 1 $\mathbb{Q} H^*(\mathbb{P}^2)$.

$$H^*(\mathbb{P}^2) = \mathbb{C}[H]/(H^3)$$

$$\text{Let } \phi_0 = 1 \quad \phi_1 = H \quad \phi_2 = H^2$$

$$\eta = \begin{pmatrix} 1 & & \\ & 1 & \\ & & 1 \end{pmatrix} \quad \eta_{ij} = \delta_{i, 2-i}$$

$$F = F^{\text{class}} + \sum_{d \geq 1} e^{\frac{Nd}{(3d-1)!}} t_2^{3d-1}$$

$$H * H \Big|_{x_2=0} = (H^2 + F_{111} H + F_{112} 1) \Big|_{x_2=0} = H^2$$

$$(H+H+H)\Big|_{x_2=0} = (H^2 + H)\Big|_{x_2=0} = (F_{211}H + F_{212}1)\Big|_{x_2=0} = e^{t_1}.$$

$$\text{So } \mathbb{Q}H(\mathbb{P}^4)\Big|_{x_2=0} = \mathbb{Q}[H] / (H^3 - e^{x_2})$$

consider the action of H on $1, H, H^2$

$$L_H = \begin{pmatrix} 1 & & \\ e^{t_1} & & \\ & & 1 \end{pmatrix}$$

Eigenvalues: $\omega^i e^{t_1/3}$ for $i=0,1,2$

where $\omega = e^{2\pi i/3}$ (6/5)

Example 2 smooth quintic $X \xrightarrow{5^{-1/5}} \mathbb{P}^4$ (5/5)

$$\textcircled{1} \dim M_{g,n}(X, \beta) = n$$

$$\textcircled{2} \text{ by Lefschetz hyperplane. } H^2(X) \cong H^2(\mathbb{P}^4) = \langle H \rangle$$

$$H^4(X) \cong H^4(\mathbb{P}^4) = \langle H^2 \rangle$$

$$\textcircled{3} \langle H+H, H \rangle = 5 + \sum_{d \geq 1} \sum_{n \geq 0} \frac{1}{n!} \langle H, H, H, \underbrace{xH, \dots, xH}_n \rangle_{\text{ant}}$$

notice the dual class of H has higher degree

$$= 5 + \sum_{d \geq 1} \sum_{n \geq 0} \frac{(x d)^n}{n!} N_{d,n} = 5 + \sum_{d \geq 1} n_d e^{dx} \sum_{k|d} k^3 = 5 + \sum_{d \geq 1} n_d d^3 \frac{e^x}{1-e^x}.$$

A distinguished vector E is needed to define a conformal Frobenius manifold:

Def: An Euler vector field E is:

$E \in \Gamma(M, TM)$ that satisfies:

$$\nabla(\nabla E) = 0 \quad \text{and}$$

$$\mathcal{L}_E(g) = (2-d) \cdot g \quad \mathcal{L}_E(\star) = \star \quad \text{for some } d$$

$$\mathcal{L}_E(a \cdot b) = (\mathcal{L}_E a) \cdot b + a \cdot \mathcal{L}_E b = (2-d) \langle a, b \rangle$$

$$\mathcal{L}_E(a \cdot b) = \mathcal{L}_E(a) \cdot b + a \cdot \mathcal{L}_E(b) = a \cdot b$$

rmk1 we further require that ∇E is diagonalizable

in flat coordinates. $(\nabla E) \partial_\alpha = (1 - g_\alpha) \partial_\alpha$

rmk2 one can get from the definition, that:

$$E(F) = (3-d)F + \text{quadratics}$$

Prop. Take $(\oplus H^i(X), \star, \tau, \epsilon)$ choose homogenous basis

$\phi_0, \dots, \phi_n$ with coordinates $t^0, \dots, t^n$.

take $d_i = \deg(\phi_i)$

$$A = \{a: \phi_a \in H^2(X)\} \quad B = \{b: \phi_b \in H^2(X)\}$$

$$\text{Let } \zeta_1(T_X) = \sum_{a \in A} z^a \phi_a$$

$$E = \sum_{a \in A} z^a \frac{\partial}{\partial t^a} - \frac{1}{2} \sum_{b \in B} (d_b - 2) t^b \frac{\partial}{\partial t^b}$$

then E is an Euler vector field with

$$d = \dim_{\mathbb{C}} X$$

Proof. $\nabla_{\partial_\alpha} E = -\frac{1}{2} (d_\alpha - 2) \frac{\partial}{\partial t^\alpha}$

$$\begin{aligned} (\mathcal{L}_E \eta)_{\alpha\beta} &= (\partial_\alpha E^\epsilon) \eta_{\epsilon\beta} + (\partial_\beta E^\epsilon) \eta_{\alpha\epsilon} \\ &= \frac{1}{2} [2 - d_\alpha + 2 - d_\beta] \eta_{\alpha\beta} = (2-d) \eta_{\alpha\beta} \end{aligned}$$

$$(\eta_{\alpha\beta} = 0 \text{ only if } d_\alpha + d_\beta = 2d$$

in general.

$$(\phi_0^{n_0} \cdots \phi_n^{n_n}) = 0 \text{ only if } \chi(\Sigma, f^* T_X) = \int_B \zeta_1(T_X) + \dim X (1-g)$$

$$\sum n_i d_i = 2 \dim \bar{M}_{0,n}(X, \beta) \quad \dim M_{g,n} = 3g-3+n$$

$$= 2 \left(\int_B \zeta_1(T_X) + (\dim X - 3)(1-g) + n \right)$$

let $\tau = \tau' + \tau''$ where $\tau' \in H^2$ $\tau'' \notin H^2$

$$\text{then } c_{\alpha\beta\gamma} = \frac{1}{d} e^{d\tau'} \sum_{n \geq 0} \frac{1}{n!} \langle \phi_\alpha \phi_\beta \phi_\gamma \tau'' \dots \tau'' \rangle_{0,n}$$

from dimension constrain.

$$\mathcal{L}_E c_{\alpha\beta\gamma} = (g_\alpha + g_\beta + g_\gamma - \dim_c X) c_{\alpha\beta\gamma} \quad \#$$

Thm Let $u_1, \dots, u_n$ be eigenvalues of E .

$$\det(U(t) - \lambda \cdot 1) = (-1)^n \prod_{i=1}^n (\lambda - u_i(t))$$

Near a semisimple point they can serve as

local coordinates. such that:

$$\frac{\partial}{\partial u_i} \cdot \frac{\partial}{\partial u_j} = \delta_{ij} \frac{\partial}{\partial u_i} \quad (1)$$

$$e_i = \sum \frac{\partial}{\partial u_i} \quad (2)$$

$$E = u_i \frac{\partial}{\partial u_i} \quad (3)$$

$$\langle \cdot, \cdot \rangle = \sum \eta_{ij}(u) du_i^2 \quad (4)$$

$$\eta_{ii} = \frac{\partial \mathcal{L}}{\partial u_i}$$

Sketch of proof:

1. show that $\exists u_i$ such that $\frac{\partial}{\partial u_i} \cdot \frac{\partial}{\partial u_j} = \delta_{ij} \frac{\partial}{\partial u_i}$
this requires showing the idempotent frames π_i are
integrable $[\pi_i, \pi_j] = 0$, which follows from flatness
of $\nabla^{(2)}$ (C1) + (C2)

2. using homogenous properties of $\mathcal{L}_E$
to get $\frac{\partial E^i}{\partial u_j} = \delta_{ij}$ (C3)

3. $\langle e, \theta \rangle = dt, \theta \rangle = \partial_u t,$

$$\langle e, \frac{\partial}{\partial u_i} \rangle = \langle e, \pi_i \cdot \bar{\pi}_i \rangle = \langle \pi_i, \bar{\pi}_i \rangle$$

Definition. coordinates satisfying (C1)-(C4) are
called canonical coordinates.

Part II Extended deformed connection.

Given $(M, \eta, \star, \ell)$

$$\Rightarrow \nabla_X^{(z)} = \nabla_X^{LC} + z X \star$$

Lemma $\nabla^{(z)}$ is flat for all z iff

$\star$ is associative + $(\alpha \beta \gamma)(t) = \frac{\partial \partial \partial F(t)}{\partial \beta \partial \gamma}$ for some f .

Pf: Take flat coordinates of ∇_X^{LC}

$$\text{then } \nabla_{\alpha}^{(z)} \partial_{\beta} = z L_{\alpha \beta}^{\sigma} \partial_{\sigma}$$

flatness of $\nabla^{(z)}$

$$\Leftrightarrow [\nabla_{\alpha}^{(z)}, \nabla_{\beta}^{(z)}] \partial_{\epsilon} = \nabla_{\alpha}^{(z)} \nabla_{\beta}^{(z')} \partial_{\epsilon} - \nabla_{\beta}^{(z')} \nabla_{\alpha}^{(z)} \partial_{\epsilon}$$

$$= \nabla_{\alpha}^{(z')} \left(z (L_{\beta \epsilon}^{\sigma} \partial_{\sigma}) \right) - \nabla_{\beta}^{(z')} \left(z (L_{\alpha \epsilon}^{\sigma} \partial_{\sigma}) \right)$$

$$= z \left[(\partial_{\alpha} (L_{\beta \epsilon}^{\sigma})) \partial_{\sigma} + (L_{\beta \epsilon}^{\sigma} z (L_{\alpha \sigma}^{\sigma'} \partial_{\sigma'})) \right]$$

$$- z \left[(\partial_{\beta} (L_{\alpha \epsilon}^{\sigma})) \partial_{\sigma} + (L_{\alpha \epsilon}^{\sigma} z (L_{\beta \sigma}^{\sigma'} \partial_{\sigma'})) \right]$$

$$= e \left(\partial_\alpha (\rho^\sigma_\epsilon - \partial_\beta (\alpha^\sigma_\epsilon)) \partial_\sigma \right) \quad (I)$$

$$+ e^2 \sum_\sigma \left(\rho^\sigma_\epsilon (\alpha^{\sigma'}_{\sigma'} - \alpha^\sigma_\epsilon (\rho^{\sigma'}_\sigma)) \partial_{\sigma'} \right) \quad (II)$$

(I) ∇_ω is totally symmetric

$\Rightarrow \exists$ pre potential

(II) associativity: $(\partial_\beta \partial_\epsilon) \partial_\alpha = (\partial_\alpha \partial_\epsilon) \partial_\beta$

rmk this provides an alternative way to define a Frobenius manifold through a pencil of flat connections.

Now suppose M has an Euler vector field E

then $\mathcal{U}(Y) := E \star Y$ and $\mu(Y) := \frac{2-d}{2} Y - \nabla_Y E$

defines (1,1) tensor on M

Defn. the extended deformed connection $\hat{\nabla}_X$ on $M \times \mathbb{C}$ is defined as:

$$\hat{\nabla}_X Y = \nabla_X^{\mathbb{C}} Y + Z X^* Y$$

$$\hat{\nabla}_{\partial_z} Y = \nabla_{\partial_z} Y + U(Y) - \frac{1}{2} \mu(Y).$$

using $\nabla E = \text{diag} \{1 - \eta_1, \dots, 1 - \eta_n\}$

$$M_\alpha = \eta_\alpha - \frac{d}{2}$$

Lemma 2 the system:

$$\begin{cases} \partial_\alpha \bar{z}_\beta = Z C_{\alpha\beta}^{\bar{\sigma}} \bar{z}_\sigma & (*) \end{cases}$$

$$\begin{cases} \partial_z \bar{z}_\beta = E^\sigma C_{\sigma\beta}^\alpha \bar{z}_\alpha - \frac{1}{2} M_\beta^e \bar{z}_e & (**) \end{cases}$$

are compatible.

$$\left(\mu \bar{z}_\beta = M_\beta^e \bar{z}_e \right)$$

Proof. $(*)$ is proved in Lemma 1.

$(**)$: we need to show $[\partial_z, \partial_\alpha] = 0$.

which can be checked to be implied by $\nabla E = \text{diag} \{1 - \eta_1, \dots, 1 - \eta_n\}$ and $\star$ is quasi-homogeneous.

Appendix L

$$\partial_z \partial_\alpha \tilde{z}_\beta = \partial_z (z (\alpha^\sigma \tilde{z}_\sigma) = (\alpha^\sigma \tilde{z}_\sigma + z (\alpha^\sigma (\partial_z \tilde{z}_\sigma))$$

$$= (\alpha^\sigma \tilde{z}_\sigma + z (\alpha^\sigma [E^\epsilon (\epsilon^\delta_\sigma \tilde{z}_\delta - \frac{1}{z} M^\eta_\sigma \tilde{z}_\eta)])$$

$$\partial_\alpha (\partial_z \tilde{z}_\beta) = \partial_\alpha (E^\sigma (\epsilon^\delta_{\sigma\beta} \tilde{z}_\delta - \frac{1}{z} M^\epsilon_{\beta\epsilon} \tilde{z}_\epsilon))$$

$$= (\partial_\alpha E^\sigma (\epsilon^\delta_{\sigma\beta} + E^\sigma \partial_\alpha (\epsilon^\delta_{\sigma\beta})) \tilde{z}_\delta$$

$$+ E^\sigma (\epsilon^\delta_{\sigma\beta} (z (\alpha^\eta_{\delta\eta} \tilde{z}_\eta))$$

$$- \frac{1}{z} (\partial_\alpha M^\epsilon_{\beta\epsilon}) \tilde{z}_\epsilon - \frac{1}{z} M^\epsilon_{\beta\epsilon} \cdot z (\alpha^\epsilon_{\alpha\epsilon} \tilde{z}_\epsilon)$$

-

$$\frac{1}{z} \text{ term: } \partial_\alpha M^\epsilon_{\beta\epsilon} = 0 \Leftrightarrow \nabla \nabla E = 0$$

$$\text{by assumption } \nabla E = \text{diag}\{1-\eta_1, \dots, 1-\eta_n\} \quad \checkmark$$

$$z \text{ term: } \sum_{\sigma, \epsilon} \left((\alpha^\sigma \partial_\alpha E^\epsilon (\epsilon^\eta_{\sigma\epsilon} - E^\epsilon (\epsilon^\sigma_{\epsilon\beta} (\alpha^\eta_{\alpha\sigma}))) \tilde{z}_\eta = 0 \right.$$

σ, ϵ

$$\text{this is commutativity: } \partial_\alpha \cdot \partial_\beta \cdot E \cdot \partial_\sigma = E \cdot \partial_\beta \cdot \partial_\alpha \cdot \partial_\sigma \quad \checkmark$$

constant term:

Appendix II

$$C_{\alpha\beta}^{\sigma} \tilde{z}_{\sigma} + C_{\alpha\beta}^{\sigma} M_{\sigma}^{\gamma} \tilde{z}_{\gamma}$$

$$- \partial_{\alpha} E^{\sigma} C_{\sigma\beta}^{\delta} - E^{\sigma} \partial_{\alpha} C_{\sigma\beta}^{\delta} \tilde{z}_{\delta}$$

$$- M_{\beta}^{\epsilon} C_{\alpha\epsilon}^{\gamma} \tilde{z}_{\gamma}$$

using $\nabla E = \text{diag}(1-q_1, \dots, 1-q_n)$

$$M = \frac{2-d}{2} - \text{diag}(1-q_1, \dots, 1-q_n)$$

$$= C_{\alpha\beta}^{\sigma} \tilde{z}_{\sigma} + C_{\alpha\beta}^{\sigma} M_{\sigma}^{\gamma} \tilde{z}_{\gamma} - M_{\beta}^{\epsilon} C_{\alpha\epsilon}^{\gamma} \tilde{z}_{\gamma} \\ - \partial_{\alpha} E^{\sigma} C_{\sigma\beta}^{\delta} - E^{\sigma} \partial_{\alpha} C_{\sigma\beta}^{\delta} \tilde{z}_{\delta} = 0$$

$$C_{\alpha\beta}^{\sigma} (1 + M_{\sigma}^{\gamma} - M_{\beta}^{\epsilon} - \partial_{\alpha} E^{\sigma})$$

$$= C_{\alpha\beta}^{\sigma} (q_{\alpha} + \delta_{\beta}^{\sigma} - q_{\sigma})$$

$$= E^{\epsilon} \partial_{\alpha} C_{\epsilon\beta}^{\sigma} = E^{\epsilon} \partial_{\epsilon} C_{\alpha\beta}^{\sigma}$$

totally symmetric

#

Now we add semi-simplicity.

Let $\pi_i = \frac{\partial}{\partial u_i}$ be idempotents.

$\rho_\alpha = \frac{\partial}{\partial t_\alpha}$ flat coordinates

introduce normalized idempotents:

$$f_i = \frac{\pi_i}{\sum \pi_i \pi_i}$$

$$\text{Let } e_\alpha = \sum \psi_{i\alpha} f_i$$

Lemma after the gauge transformation:

$$y = \Psi z$$

the system $(*)$ and $(**)$ in $\{u_i, z\}$ chart reads

$$\begin{cases} \partial_i y = (z E_i + V_i) y & (*) \\ \partial_z y = (U + \frac{V}{z}) y & (**) \end{cases}$$

$$\text{where: } (E_i)_{ab} = s_{ia} s_{ib} \quad V_i = \partial_i \Psi \cdot (\Psi^{-1})$$

V_i is uniquely determined by $\begin{cases} \text{skew symmetric} \\ [U, V_i] = [E_i, V] \end{cases}$

$$U = \text{diag}(u_1, \dots, u_n)$$

Moreover. V satisfy: $\partial_i V = [V_i, V]$

rmk. fix $u_1, \dots, u_n$

then $(**)$ becomes a connection on $\begin{array}{c} \mathbb{P}^1 \times T_u M \\ \downarrow \\ \mathbb{P}^1 \end{array}$

which has regular singularity at $z = 0$

and irregular singularity at $z = \infty$

$$\left(z = \frac{1}{\omega} \quad \partial_{\omega} y = -\frac{1}{\omega} \partial_z y = -\frac{1}{\omega} (U + \omega V) y = -\left(\frac{U}{\omega} + V\right) y \right)$$

$\partial_i V = [V, V]$ is a isomonodromic condition, in u